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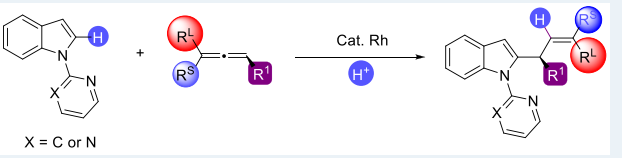
Rhodium-Catalyzed Intermolecular Stereoselective Allylation of Indoles with Allenes

Chaoyan Ma, Can Li, Jiahui Bai, Junzhe Xiao, Yizhan Zhai,* Yinlong Guo,* and Shengming Ma*

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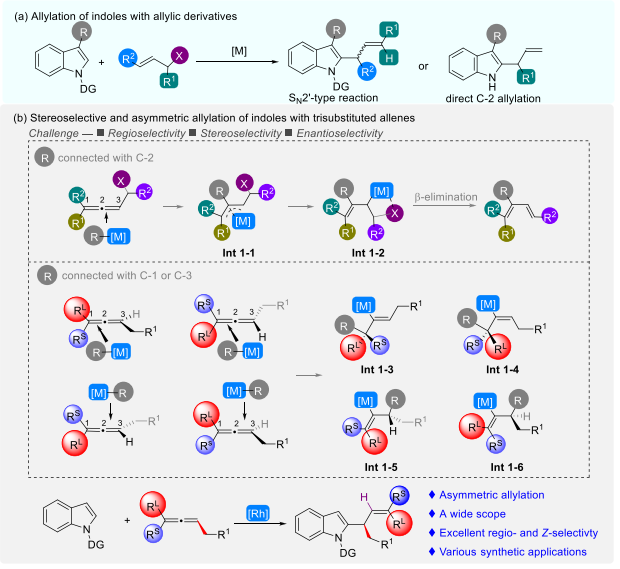
ABSTRACT: A rhodium-catalyzed C−H functionalization-based intermolecular allylation of indoles with trisubstituted allenes has been developed. This synthetic process proceeds with an excellent regioselectivity, *Z*-selectivity, high efficiency of chirality transfer, and a wide tolerance of reactive functional groups under mild conditions. Late-stage modification on a series of complex bioactive or drug molecules and various synthetic transformations has been demonstrated. A reaction mechanism has been proposed on the basis of mechanistic experiments and SAESI-MS studies.



KEYWORDS: *indoles, allenes, allylation, rhodium-catalysis, chirality transfer, Z-selectivity*

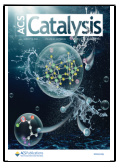
Indoles are intriguing and fundamental structural motifs in numerous natural products and pharmaceuticals.[1] With different transformations of indoles, new C−C and C−X bonds could be selectively constructed at 2, 3, or 7 positions,[2] providing an important route to diverse structures. 2-Allylated indoles are of current interest due to the synthetic and biological potentials.[3] However, there is a challenge for the allylation at the C-2 position of indoles: it could be completed by S_N2'-type reactions with allylic derivatives with *E*/*Z* stereoselectivity unsolved or direct C-2 allylation (Scheme 1a).[4,5] In addition, no asymmetric allylation has been reported with the indole derivatives. We reasoned that allenes may be applied for such a purpose. In principle, allenes would react with R−M mostly to afford the π-allylic metal intermediate **Int 1−1** (Scheme 1b, top).[6] For allylation reaction with allenes, however, the regioselectivity of the insertion step has to be reversed to the noncommon way to produce alkenyl metal intermediates **Int 1−3 ∼ Int 1−6** (Scheme 1b, middle).[6b,c,7] So far, such a regioselectivity switch has not been observed with the indole derivatives. The challenges would be (1) selective formations of **Int 1−3 ∼ Int 1−6**; (2) the protodemetalation of the C−M bonds in **Int 1−3 ∼ Int 1−6**; and (3) the stereoselectivity referring to the remaining C=C bond and chirality transfer of optically active allenes fulfilling the allene-based asymmetric allylation. Here, we wish to report rhodium-catalyzed asymmetric allylation of indoles with trisubstituted allenes (Scheme 1b, bottom).

Scheme 1. Transition Metal-Catalyzed C−H Allylation of Indoles



We began our exploration by evaluating the allylation reaction between indole derivatives **1a-d** with different directing groups and allene **2a**[8b] in the presence of $[(Cp^*RhCl_2)_2]$, $AgSbF_6$, and PivOH in toluene: indoles with Piv (**1a**) and *t*-Bu_2P (**1b**) failed to react while indoles with 2-pyridinyl (**1c**) and 2-pyrimidinyl (**1d**), as the directing group could react smoothly to afford allylation products **3ca** and **3da**, which means the C2 carbon of indole has been connected to the C2 carbon of the allene unit (Table 1, entries 1−4). Regioisomeric product 1,3-diene **4** via β-O elimination was not detected. The reaction conditions were optimized with 2-pyrimidinyl as the directing group: solvent

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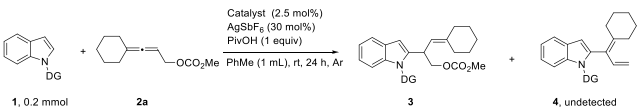


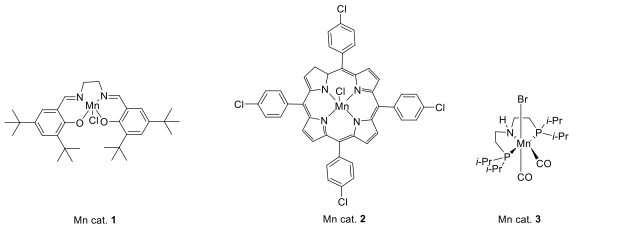
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Table 1. Optimization of Reaction Conditions for the Reaction of 1 with 2a[a]





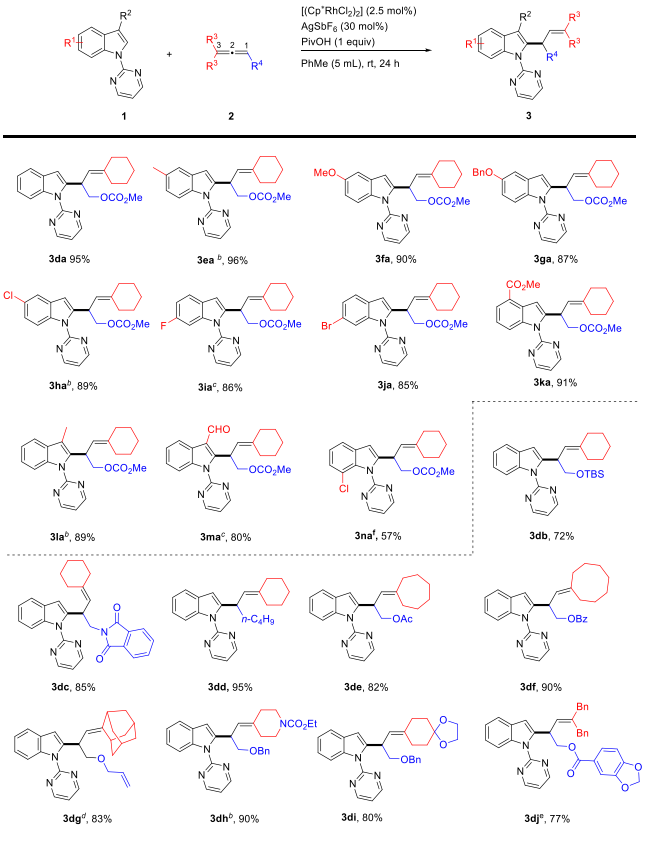
entry	DG	catalyst	NMR recovery of **1** (%)	NMR yield of **3** (%)
1	Piv (**1a**)	$[(Cp^*RhCl_2)_2]$	80	–
2	*t*-Bu_2P (**1b**)	$[(Cp^*RhCl_2)_2]$	93	–
3[b]	2-pyridinyl (**1c**)	$[(Cp^*RhCl_2)_2]$	–	**3ca** 88(83[c])
4	2-pyrimidinyl (**1d**)	$[(Cp^*RhCl_2)_2]$	–	**3da** 96
5[c]	(**1d**)	$[(Cp^*RhCl_2)_2]$	74	**3da** 17
6[d]	(**1d**)	$[(Cp^*RhCl_2)_2]$	5	**3da** 76
7	(**1d**)	$RhCl_3$	75	–
8[e]	(**1d**)	$[(Cp^*RhCl_2)_2]$	89	–
9[g]	(**1d**)	$[(Cp^*RhCl_2)_2]$	80	**3da** 15
10[h]	(**1d**)	$[(Cp^*RhCl_2)_2]$	–	**3da** 93
11[i]	(**1d**)	$[(Cp^*RhCl_2)_2]$	–	**3da** 93
12	(**1d**)	Cp^*RuCl_2	93	–
13	(**1d**)	$[(Cp^*IrCl_2)_2]$	94	**3da** < 1
14	(**1d**)	$Cp^*CoI_2(CO)$	76	**3da** 10
15	(**1d**)	$Mn(CO)_5Br$	86	**3da**
16	(**1d**)	Mn cat. **1**	92	–
17	(**1d**)	Mn cat. **2**	91	–
18	(**1d**)	Mn cat. **3**	99	–

[a]The reactions were carried out using 0.2 mmol of **1**, 1.1 equiv of **2a**, 2.5 mol % of $[Cp^*RhCl_2]_2$, 30 mol % of $AgSbF_6$, and 1 equiv of PivOH in 1 mL of PhMe at room temperature. [b]The reaction was carried out on 0.5 mmol scale. [c]Isolated yield. [d]CH_3CN as solvent. [e]DMF as solvent. [f]No $AgSbF_6$. [g]No PivOH. [h]20 mol % $AgSbF_6$. [i]10 mol % $AgSbF_6$

screening revealed that the reaction in CH_3CN was very slow, and the reaction in DMF was incomplete (Table 1, entries 5 and 6). $RhCl_3$ failed to catalyze the reaction (Table 1, entry 7); when the reaction was conducted in the absence of $AgSbF_6$ or PivOH, no or a very minor amount of **3da** was formed, respectively (Table 1, entries 8 and 9). The loading of $AgSbF_6$ may also be reduced to 10 mol % (Table 1, entries 10 and 11). As for other transition metals such as [Ru], [Ir], [Co], and [Mn] catalysts, the reaction proceeded sluggishly or even did not work (Table 1, entries 12−18). Therefore, $[(Cp^*RhCl_2)_2]$ is an optimal catalyst for the reaction system.

To evaluate the generality of this reaction, we began to explore the scope of trisubstituted allenes **2** with indole derivatives **1** (Table 2). First, the scope of indoles was demonstrated with 4,4-pentamethylene-2,3-butadienyl methyl carbonate **2a**: in addition to the inert 5-Me and 3-Me, synthetically versatile 5-MeO, 5-BnO, 5-Cl, 6-F, 6-Br, 4-CO_2Me, 3-CHO, and 7-Cl could be tolerated regardless of the electronic difference affording the corresponding products **3ea**−**3na**. Notably, the indoles with 3-

Table 2. Reaction Scopes



[a]Reaction conditions: The reactions were carried out using 1 mmol of **1**, 1.1 equiv of **2**, 2.5 mol % of $[Cp^*RhCl_2]_2$, 30 mol % of $AgSbF_6$, 1 equiv of PivOH, and 5 mL of toluene at room temperature. Yields of isolated products were given. [b]The reaction was run for 12 h. [c]The reaction was carried out at 50 °C. [d]The reaction was run for 8 h. [e]The reaction was carried out at 70 °C. [f]9% of **3na′** product (for its structrue see SI) was obtained.

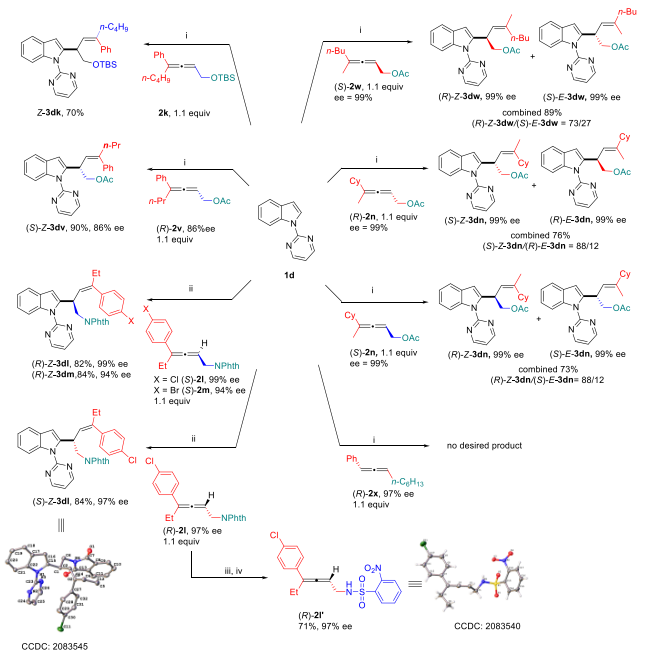
position substitution of 3-Me (**1l**) and 3-CHO (**1m**) increasing the steric hindrance also worked. The scope of trisubstituted allenes with various functional groups was also tested: the 3-position of the allenes may be substituted with a six- to eight-membered ring and adamantine. TBS ether, phthalimide, *n*-butyl, acetate, benzoate, and an allyl ether group may all be accommodated (**3db**−**3dg**). The reaction of a substrate bearing acid-sensitive functional groups such as cyclic carbamate **2h** and ketal **2i** also worked smoothly under the standard reaction conditions, providing related products **3dh** and **3di**. The reaction could be extended to the acyclic system, such as 4,4-dibenzyl substitution together with piperonate unit **2j**, giving the corresponding **3dj** at 70 °C.

Furthermore, the reactions with nonsymmetric trisubstituted and optically active allenes were carried out under the standard conditions (Scheme 2). Nonsymmetric allene **2k** afforded *Z*-**3dk** with an excellent *Z*-stereoselectivity. In addition, when the optically active allenes (*R*)-**2v**, (*S*)-**2l** (*S*)-**2m**, and (*R*)-**2l** were employed, high efficiency of chirality transfer could be achieved, giving (*S*)-*Z*-**3dv**, (*R*)-*Z*-**3dl**, (*R*)-*Z*-**3dm**, and (*S*)-*Z*-**3dl** with retention of the ee value in decent yields and an excellent *Z* selectivity. The absolute configurations of (*S*)-*Z*-**3dl** and (*R*)-**2l′** have been established by X-ray diffraction study, which demonstrated the steric outcome of this reaction. Similarly,

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Scheme 2. *Z* Selectivity and Axial Chirality Transfer Reactions[a]

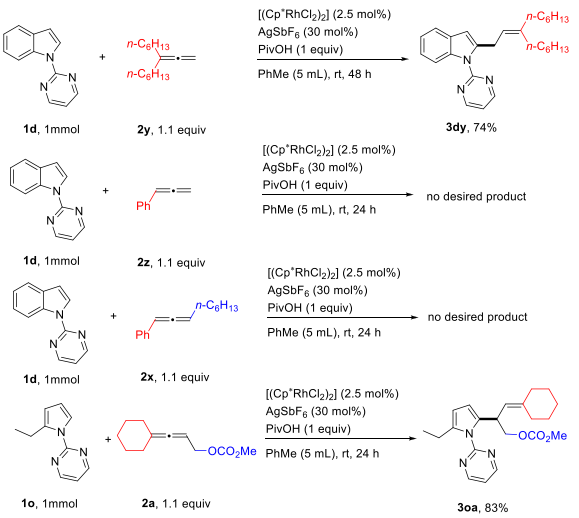


[a]Reagents and conditions: (i) $[Cp^*RhCl_2]_2$ (2.5 mol %), $AgSbF_6$ (30 mol %), PivOH (1 equiv), toluene, rt. (ii) $[Cp^*RhCl_2]_2$ (2.5 mol %), $AgSbF_6$ (30 mol %), PivOH (1 equiv), toluene, 70 °C. (iii) $N_2H_4 \cdot H_2O$ (6 equiv), MeOH, rt. (iv) 2-Nitrobenzenesulfonyl chloride (1 equiv), Et_3N (2 equiv), DCM, rt.

when the optically active compounds (*S*)-**2w** (99% ee), (*R*)-**2n** (99% ee), and (*S*)-**2n** (99% ee) were applied, (*R*)-*Z*-**3dw** (99% ee), (*S*)-*Z*-**3dn** (99% ee), and (*R*)-*Z*-**3dn** (99% ee) could be obtained, respectively, with a *Z*/*E*-selectivity of 73:27 to 88:12 and an excellent level of efficient chirality transfer. In addition, employing optically active 1,3-disubstituted allene (*R*)-**2x** could not afford the desired product.

Moreover, some other type substrates were also attempted (Scheme 3): the reaction of 1,1-disubstituted allene **2y** could afford **3dy** in a yield of 74%, whereas no allylation product could

Scheme 3. Attempt for the Reactions of Other Type Allene Substrates



be detected for the monosubstituted allene **2z** and 1,3-disubstituted allene **2x**. To our delight, when pyrrole derivative **1o** was applied with **2a**, it also successfully gave the desired product **3oa** in 83% yield.

To demonstrate the synthetic utility and practicality of the method, a series of synthetic applications was carried out. Firstly, a gram-scale reaction of 3 mmol scale was conducted, which afforded 1.25 g (91% yield) of **3ho** (Scheme 4a). Next, trisubstituted allenes bearing units from natural products or drug molecules were synthesized according to the ATA[7b] reaction and esterification. Under the standard conditions, a range of cyclic and acyclic trisubstituted allenes derivatized from (1*S*)-(−)-camphanicacid (**5**), adapalene (**6**), cholesteryl hemisuccinate (**7**), indometacin (**8**), fenofibrate acid (**9**), isoxepac (**10**), oxaprozin (**17**), and febuxostat (**18**) are compatible, affording the related products **3dp**−**3dB** in decent yields and complete *Z*-stereoselectivity for compound **3du** (Scheme 4b). Furthermore, such products are synthetically versatile: when indole product **3dd** was treated with NaOEt and DMSO at 110 °C, the directing group could be removed to generate corresponding indole **11**.[9] Next, homoallyl alcohol compound **12** could be yielded by reacting with TBAF·$3H_2O$; compound **13** could be formed with phenylacetylene via Pd-catalyzed Sonogashira coupling reaction.[10] 2-Pyrimidinyl may be further applied to implement C−H functionalization at C-7 of indole with methyl acrylate to afford compound **14**.[11] Suzuki coupling reaction of (*R*)-**3dm** with $PhB(OH)_2$ could generate compound **15**[12] without racemization. Phthalimide could be removed and the subsequent reaction with tosyl chloride afforded tosyl amide **16** without racemization (Scheme 4c).

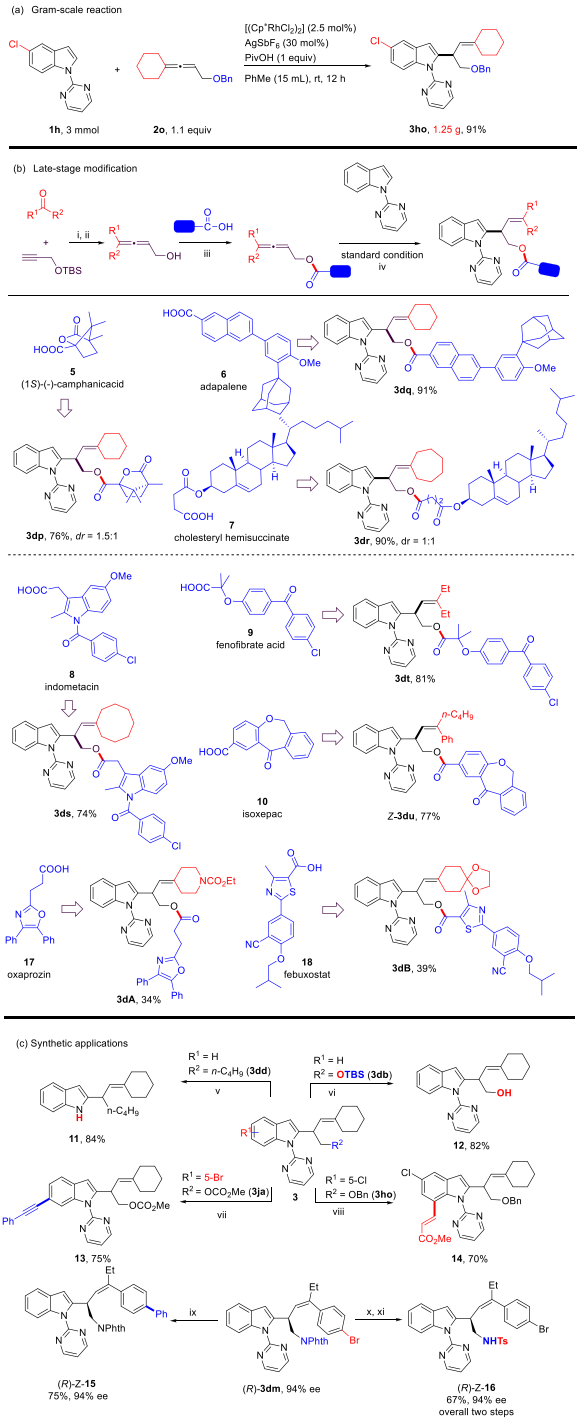
In order to gain insight into the reaction mechanism, a series of deuterium-labeling and control experiments were conducted (Scheme 5). First, approximately 26% deuterium incorporation was observed at the 2-position of **1d** when the reaction of **1d** was conducted with 1 equiv of PivOD in the absence of **2a**, indicating that the C−H bond activation process is reversible (Scheme 5, eq 1). Next, the reaction of **1d** with **2a** was conducted with 1 equiv of PivOD under the standard conditions; a 38% deuterium incorporation onto the olefinic position of product **3da** was observed (Scheme 5, eq 2). Moreover, under otherwise identical conditions, the reaction employing **1d-D1** with D incorporation of 96% at the 2-position with **2a** afforded the product **3da-D1** with 56% and 20% deuterium incorporation at the vinylic position 7-position of the indole moiety, respectively (Scheme 5, eq 3). A control experiment using **1d-D1** and PivOH led to a decrease of deuterium incorporation at the vinylic position (Scheme 5, eq 4). These experiments indicated that olefinic hydrogen originated from the proton of PivOH and the 2-position of **1d**. The kinetic isotope effect was determined by using **1d** and **1d-D1** via parallel and competitive experiments to be 1.00 and 1.15 based on the integration of indole C3 H NMR analysis, indicating that the C−H activation is not the rate-determining step (Scheme 5, eqs 5−7). Moreover, to verify the possibility for C-7 allylation, two experiments were conducted: when increasing the amount of **2a** and prolonging the reaction time, the allylation at C-7 could not be achieved (Scheme 5, eq 8); the reaction of product **3da** with 1.1 equiv. of **2a** under the standard conditions still failed to afford the product functionalized at the C-7 position (Scheme 5, eq 9). Thus, the C−H allylation occurred exclusively at the 2-position.

Furthermore, a solvent-assisted electrospray ionization mass spectrometric experiment (SAESI-MS) was conducted to detect

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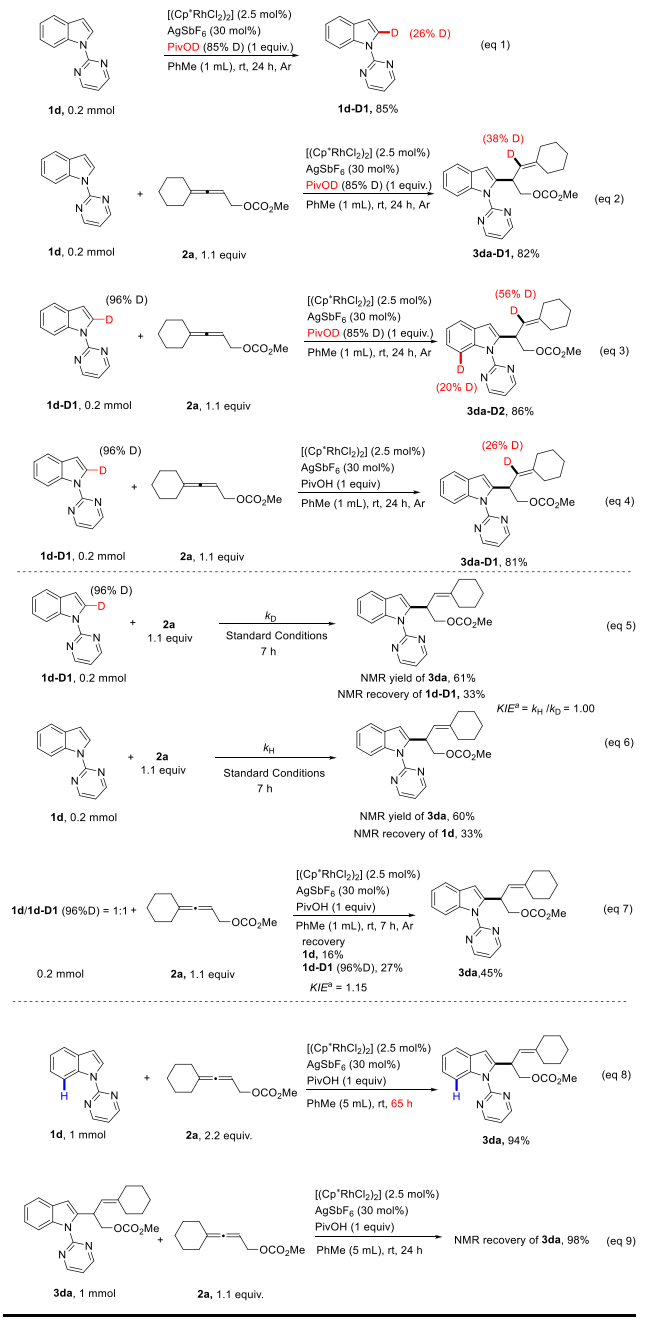
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Scheme 4. Synthetic Applications[a]



[a]Reagents and conditions: (i) Pyrrolidine (1.1 equiv), CuI (10 mol %), $ZnBr_2$ (0.8 equiv), $Ti(OEt)_4$, toluene, 120 °C. (ii) TBAF·$3H_2O$ (1.5 equiv), THF, rt. (iii) EDCI (1.2 equiv), DMAP (20 mol %), DCM, rt. (iv) $[Cp^*RhCl_2]_2$ (2.5 mol %), $AgSbF_6$ (30 mol %), PivOH (1 equiv), toluene. (v) NaOEt (4 equiv), DMSO (2 mL), 110 °C, 30 min. (vi) TBAF·$3H_2O$ (1.5 equiv), THF, rt, 12 h. (vii) Phenylacetylene (1.5 equiv), $Pd(PPh_3)_2Cl_2$ (3 mol %), PPh_3 (6 mol %), CuI (4.5 mol %), *i*-Pr_2NH, 80 °C, 18 h. (viii) Methyl acrylate (1.5 equiv), $[(Cp^*RhCl_2)_2]$ (5 mol %), $Cu(OAc)_2·H_2O$ (1 equiv), DMF (1 mL), 80 °C, 24 h. (ix) Phenylboronic acid (1.2 equiv), $Pd(PPh_3)_4$ (2 mol %), K_3PO_4 (2 equiv), DME/H_2O (4/1), 65 °C, 17 h. (x) $N_2H_4·H_2O$ (6 equiv), MeOH, rt, 12 h. (xi) TsCl (1 equiv), Et_3N (2 equiv), DCM, 24 h. Yields of isolated products were given.

Scheme 5. Mechanistic Studies



[a]KIE = (1 − recovery of **1d**)/(1 − recovery of **1d-D1**)

any active species in this reaction (Figure 1).[13] The compounds **1d** and **2a** were selected as model substrates under the standard conditions. Two signals with *m*/*z* of 432 (Figure 1a) and 710 (Figure 1b), which match the *m*/*z* of five-membered rhodacycle intermediate **MS-Int-1** (calcd for $C_{22}H_{23}N_3Rh$) and alkenyl rhodium intermediate **MS-Int-2** (calcd for $C_{37}H_{45}N_5O_3Rh$), respectively, were detected. The identities of these signals were further confirmed by the SAESI-MS/MS experiments (for details, see Supporting Information, Figure S2)

On the basis of the above experimental results, a plausible reaction mechanism was proposed (Scheme 6): Initially, Rh(III)-catalyzed reversible C−H cleavage of **1d** would form five-membered rhodacycle **Int 2−1**, which is confirmed by the detection of **MS-Int-1**, via a CMD process with the assistance of

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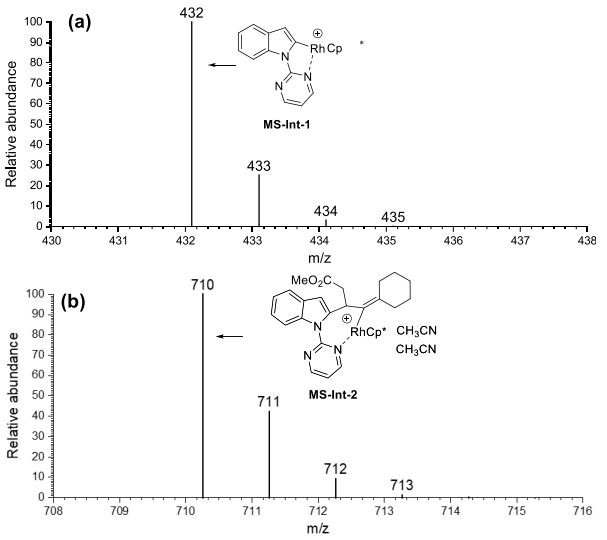
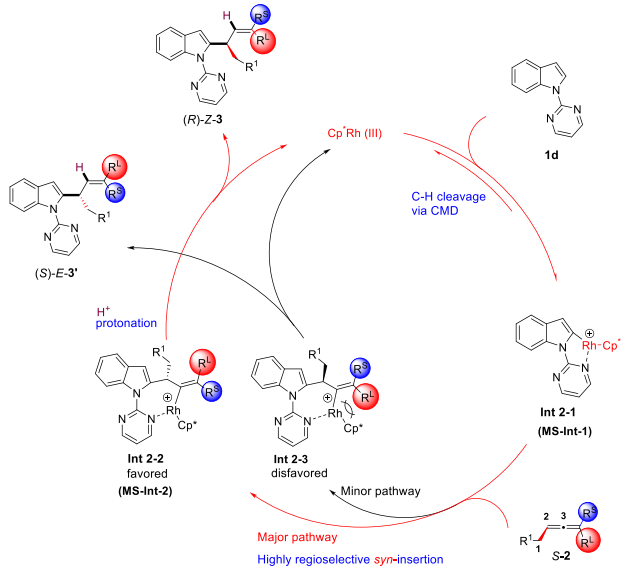


Figure 1. SAESI-MS studies. (a) Expanded SAESI-MS spectrum, showing the major signal from m/z 430 to 438. (b) Expanded SAESI-MS spectrum, showing the major signal from m/z 708 to 716.

Scheme 6. Possible Mechanism



2-pyrimidinyl as the directing group. Subsequently, trisubstituted allene **2** undergoes a highly regioselective *syn*-insertion with the C−Rh bond in **Int 2−1**, leading to the formation of **Int 2−2** supported by the detection of **MS-Int-2**.

Subsequent protonation of **Int 2−2** or **Int 2−3** afforded corresponding product (*R*)-*Z*-**3** or (*S*)-*E*-**3′**, followed by releasing the rhodium catalyst to finish the catalytic cycle. Obviously, the *Z*-stereoselectiovity may be rationalized by the fact that **Int 2−3** is disfavored owing to steric hindrance. The efficiency of chirality transfer may be explained by the strict *syn*-insertion and stereospecific protodemetaltion.

In conclusion, we have developed a novel rhodium-catalyzed asymmetric C−H allylation of indoles with trisubstituted allenes under mild conditions. The overall process exhibits excellent atom economy, unique regioselectivity and stereoselectivity, and good functional group tolerance. Synthetic applications as well as the mechanistic investigation have been conducted and enhanced the utility and understanding of the reaction.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.2c01433.

Experimental procedures, compound characterization data, SAESI-MS data, and NMR and HPLC spectra (PDF)

Crystallographic data for (*R*)-**2l′** (CIF)

Crystallographic data for (*S*)-*Z*-**3dl** (CIF)

■ AUTHOR INFORMATION

Corresponding Authors

Yizhan Zhai − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China*; Email: zhaiyz@sioc.ac.cn

Yinlong Guo − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China*; Email: ylguo@sioc.ac.cn

Shengming Ma − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China; Laboratory of Molecular Recognition and Synthesis, Department of Chemistry, Zhejiang University, Hangzhou, Zhejiang 310027, People's Republic of China; Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, People's Republic of China*; orcid.org/0000-0002-2866-2431; Email: masm@sioc.ac.cn

Authors

Chaoyan Ma − *Center of Chemistry for Frontier Technologies, Department of Chemistry, Zhejiang University, Hangzhou, Zhejiang 310027, People's Republic of China*

Can Li − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China*

Jiahui Bai − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China*

Junzhe Xiao − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.2c01433

Notes

The authors declare no competing financial interest.

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